

LOW MOLECULAR WEIGHT NBR: PATENT LANDSCAPE FOR CONTROLLED SOLUTION AND PLASMA-POLYMER PROCESSES

1. ABSTRACT

This report reviews patents relevant to low molecular weight nitrile butadiene rubber (NBR), with emphasis on synthesis routes that deliberately reduce chain length, lower viscosity, or create functional low-MW intermediates. The patent set spans two distinct technical directions. EP2478021B1 discloses solution free-radical polymerization of butadiene and acrylonitrile in organic solvent using a modifier/regulator system, with explicit adjustment of regulator concentration to obtain low molecular weight NBR suitable for downstream RAFT-based block/terpolymer construction. The examples show both conventional NBR preparation and deliberate low-MW product formation, including a reported M_n of 12,000 g/mol for a RAFT macromer. US10400053B2 is not a synthesis patent for NBR itself, but it is relevant to low-MW NBR compounding because it describes plasma polymerization of hydrocarbon monomers on carbon particles to create active fillers for rubber blends, including NBR-containing formulations. Together, the patents illustrate the broader process landscape around low-viscosity NBR: molecular-weight control by chain-transfer/regulator chemistry in solution polymerization, conversion control and staged monomer addition, and downstream filler-surface engineering to improve processability and composite performance. Across the evidence, the dominant polymerization themes are radical solution polymerization in organic solvent, use of RAFT/regulator chemistry, oxygen exclusion, butadiene pressurized feed, elevated temperature initiation, and solvent-based isolation by precipitation. The broader formulation/process trend is the move away from conventional emulsion-polymerized NBR toward more controllable organic-solution routes that can generate lower-MW, functionalized, or block-capable NBR intermediates while preserving compatibility with later blending or hydrogenation steps.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2478021B1

- Patent Title: Nitrile rubbers and production of same in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses free-radical solution polymerization of at least one conjugated diene and at least one alpha,beta-unsaturated nitrile in an organic solvent in the presence of a modifier/regulator. The process is used to prepare nitrile rubbers with controlled molecular weight, including deliberately low molecular weight NBR, and can optionally be followed by hydrogenation. - Vazo® 88: 284.9 mg — Example 1 - Vazo® 88: 1.17 mmol, 0.31 phm — Example 1 - DoPAT: 204 mg — Example 1 - DoPAT: 0.583 mmol, 0.22 phm — Example 1 - Dimethylacetamide: 50 ml, 51 phm — Example 1 - Acrylonitrile: 43 ml, 653.3 mmol, 37 phm — Example 1 - Argon degassing: 10 minutes — Example 1 - 1,3-Butadiene: 90 ml, 1077 mmol, 63 phm — Example 1 - Butadiene feed pressure: 3 bar — Example 1 - Polymerization temperature: 100°C — Example 1 - Polymerization time: 9 hours — Example 1 - Example 6 final conversion: 39% — Example 6 - Example 7 regulator amount: 2.22 phm — Example 7 - Example 7 polymerization time: 9 h — Example 7 - Example 7 final conversion: 48% — Example 7 - NBR #7 Mn: 12,000 g/mol — Example 8 - Styrene: 2.79 g, 26.8 mmol, 50 phm — Example 8 - Example 8 reaction temperature: 100°C — Example 8 - Example 8 reaction time: 13 hours — Example 8 - Example 8 final conversion: 9.2% — Example 8 - Examples 17-23 butadiene feed pressure: 3 bar — Example 17-23 - Examples 17-23 polymerization temperature: 100°C — Example 17-23 - Examples 17-23 polymerization time: 9 hours — Example 17-23 - Examples 18, 20 and 23 Vazo® 88 post-dose: 128 mg — Example 18/20/23 - Examples 18, 20 and 23 Vazo® 88 post-dose: 0.523 mmol, 0.15 phm — Example 18/20/23 - Examples 18, 20 and 23 post-dose solvent: 5 ml dimethylacetamide — Example 18/20/23

Relevant Experimental Evidence - Example 1: Baseline solution polymerization of NBR from acrylonitrile and butadiene in DMAc at 100°C and 3 bar butadiene feed. - Example 6: Terpolymerization with hydroxyethyl acrylate (HEA) as a termonomer, showing incorporation of additional functionality. - Example 7: Deliberate preparation of low molecular weight NBR by increasing regulator amount to 2.22 phm. - Example 8: Use of the low-MW NBR from Example 7 as a RAFT macromer for styrene chain extension, demonstrating block-terpolymer formation. - Example 17: Solution polymerization series with optional HDODA and HEA termonomers under the same 100°C, 3 bar regime.

Technical Relevance This is the most directly relevant patent in the set because it explicitly targets low molecular weight NBR through regulator-controlled solution polymerization. It provides concrete process levers for lowering molecular weight and generating functional NBR intermediates suitable for further chain extension or modification.

Patent 2

- Patent Number: US10400053B2
- Patent Title: Carbon particles coated with polymer films, methods for their production and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses plasma polymerization of a monomer onto carbon residue particles to produce active fillers for rubber, rather than synthesis of NBR itself. It is relevant to NBR compounding because the treated fillers are intended for use in rubber blends, including NBR-containing systems, and the process is solvent-free and conducted in plasma reactors. - RF Power: 250 watts — Example 1 - Acetylene concentration: 200 milliTorrr — Example 1 - Treatment time: 1 hr — Example 1 - RF Power: 150 watts — Example 1 (EP-P434-5) - Carbon sample BET specific surface area: 66.8 m²/g — Example 3 - Fullerenic soot sample mass: 100 g — Example 3 - Chamber pressure: 10 Pa — Example 3 - Monomer purity: 99.6% acetylene — Example 3 - RF frequency: 13.56 MHz — Example 3 - Monomer concentration: 20 Pa — Example 3 (RP-P579-1) - RF power: 80 watts — Example 3 (RP-P579-1) - Treatment time: 1 hr — Example 3 (RP-P579-1) - Monomer concentration: 40 Pa — Example 3 (RP-P579-2) - RF power: 100 watts — Example 3 (RP-P579-2) - Treatment time: 1 hr — Example 3 (RP-P579-2) - Monomer concentration: 40 Pa — Example 3 (RP-P579-3) - RF power: 100 watts — Example 3 (RP-P579-3) - Treatment time: 2 hr — Example 3 (RP-P579-3) - Monomer concentration: 50 Pa — Example 3 (RP-P579-4) - RF power: 100 watts — Example 3 (RP-P579-4) - Treatment time: 2 hr — Example 3 (RP-P579-4) - Monomer concentration: 50 Pa — Example 3 (RP-P579-5) - RF power: 100 watts — Example 3 (RP-P579-5) - Treatment time: 2.5 hr — Example 3 (RP-P579-5) - Monomer concentration: 40 Pa — Example 3 (RP-P579-6) - RF power: 100 watts — Example 3 (RP-P579-6) - Treatment time: 2.5 hr — Example 3 (RP-P579-6) - NBR Mooney viscosity: ML(1+4)100°C 45 — Example 3 - NBR acrylonitrile content: 34.7±1 wt % — Example 3 - SBR Mooney viscosity: ML(1+4)100°C 65 — Example 3 - SBR styrene content: 25 wt % — Example 3 - SBR vinyl content: 50 wt % — Example 3 - EPDM Mooney viscosity: ML(1+4)125°C 65 — Example 3 - EPDM ethylene content: 48 wt % — Example 3 - EPDM ethylidene norbornene content: 9.0 wt % — Example 3

Relevant Experimental Evidence - Example 1: Plasma coating conditions were varied across carbon types, showing measurable acetylene deposition and surface modification. - Example 2: Treated carbon black showed reduced surface energy, with 1 hour plasma treatment on fullerenic soot comparable to 4 hours on furnace carbon black. - Example 3: Plasma-coated fullerenic soot was evaluated in rubber blends, including NBR with Mooney viscosity 45, demonstrating relevance to rubber compounding. - Example 3: Process variables such as RF power, monomer pressure, and treatment time were correlated with coating weight loss. - Example 1: The patent showed that fullerenic soot can accept more uniform plasma polymer deposition than conventional carbon blacks.

Technical Relevance Although not an NBR synthesis patent, it is relevant to low-MW NBR because it addresses filler surface modification for rubber formulations where low-viscosity NBR is often used for processability. The disclosed plasma-coated fillers can influence compound rheology and performance in NBR-based blends.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2478021B1 is a direct low-MW NBR synthesis patent, while US10400053B2 is a downstream rubber-filler modification patent rather than a polymerization route.
- EP2478021B1 uses organic-solution free-radical polymerization of butadiene/acrylonitrile with a regulator/RAFT-type modifier; US10400053B2 uses solvent-free RF plasma polymerization of acetylene on carbon particles.
- EP2478021B1 explicitly controls molecular weight via regulator amount, including 2.22 phm to obtain low-MW NBR and Mn 12,000 g/mol; US10400053B2 does not disclose NBR molecular-weight control, but reports NBR compound Mooney viscosity of 45 in a rubber blend context.
- EP2478021B1 operates at 100°C and 3 bar butadiene feed with 9 h reaction times; US10400053B2 uses plasma conditions such as 250 W, 200 mTorr, 1 h, and 13.56 MHz RF frequency.
- EP2478021B1 isolates polymer by ethanol precipitation and vacuum drying; US10400053B2 characterizes coated fillers by TGA and XPS rather than polymer isolation.

4. CONCLUSION

The evidence supports solution free-radical polymerization with regulator/chain-transfer control as the most direct route to low molecular weight NBR. EP2478021B1 provides the strongest guidance: use organic solvent solution polymerization of butadiene and acrylonitrile, maintain oxygen-free conditions, feed butadiene at 3 bar, initiate at 100°C, and increase regulator loading to lower molecular weight, with a demonstrated low-MW product at Mn 12,000 g/mol. US10400053B2 is complementary rather than synthetic, showing that plasma-modified carbon fillers can be used in rubber blends, including NBR-containing systems, to tune composite properties. For low-MW NBR synthesis, the best-supported parameters are organic-solution radical polymerization, explicit regulator dosing, controlled monomer feed, and precipitation isolation; for formulation, plasma-treated fillers may help optimize processability and composite performance.

5. REFERENCES

- EP2478021B1 | Nitrile rubbers and production of same in organic solvents | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2478021B1/en>
- US10400053B2 | Carbon particles coated with polymer films, methods for their production and ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10400053B2/en>